NUPUR DASHPUTRE

nupurdashputre@gmail.com | (213) 994-7674 | Los Angeles, CA

LinkedIn | GitHub | Portfolio

EDUCATION

University of Southern California — Los Angeles, CA
Master of Science in Computer Science — August 2024 – May 2026
Related Coursework: Algorithms, Information Retrieval, Database Systems, Multimedia Systems Design, Applied NLP

MIT – World Peace University — Pune, India
Bachelor of Technology in Computer Science and Engineering — August 2020 – June 2024
Related Coursework: Data Structures, Computer Networks, Machine Learning, Web Development, Operating Systems, Software Engineering

TECHNICAL SKILLS

Languages & Systems: Python, C++, Java, JavaScript, TypeScript, SQL, Bash, Linux/Unix, Multithreading, TCP/IP
AI Development Tools: Claude, Cursor (AI-assisted code generation, code review, workflow automation)
Backend & APIs: FastAPI, Django, Node.js, REST APIs, PostgreSQL, Kafka
Cloud & DevOps: Docker, AWS, GCP, GitHub Actions (CI/CD), Prometheus, Grafana
Data & AI: PyTorch, TensorFlow, Scikit-Learn, Pandas, NumPy, LangChain, RAG, LoRA

EXPERIENCE

Sigma Healthsense Inc. — Marina del Rey, CA
Software Engineer — July 2026 – Present
- Designing **distributed** backend services and **REST APIs** using **TypeScript**, **Node.js**, **PostgreSQL**, and **GCP**, enabling reliable patient-request processing for an initial launch cohort of **300+** patients through scalable, fault-tolerant workflows.
- Building data **validation** and **orchestration** pipelines that use **Claude** to parse **voice-based patient requests** into structured clinical records, automating previously manual processing steps.

Lumina Health AI Institute (non-profit) — Los Angeles, CA
Software Engineer (Volunteer) — September 2025 – May 2026
- Designed, deployed, and maintained **25+ FastAPI** and **PostgreSQL** services on **GCP Cloud Run**, backed by **pytest** unit and integration tests in **GitHub Actions CI**, delivering **150–250 ms** response times for healthcare application data.
- Implemented observability and monitoring using **Prometheus**, **Grafana**, and structured logging, improving system visibility and **accelerating issue detection and troubleshooting**.
- Built **multilingual AI voice agents** (Retell AI) for patient onboarding and appointment scheduling, automating intake, collecting structured patient information, and **expanding healthcare access** for non-English-speaking patients.

Rolls-Royce Power Systems AG — Pune, India
Data Analyst Intern — January 2024 – June 2024
- Engineered Python data pipelines processing **20M+** industrial sensor records for ingestion, transformation, and serving, enabling scalable analytics for reliable **downstream maintenance systems**.
- Developed and deployed real-time **Django APIs** for predictive maintenance workflows, **improving data accessibility** and system responsiveness while supporting **large-scale telemetry processing**.

Persistent Systems — Pune, India
Software Intern — June 2023 – August 2023
- Developed high-performance concurrent **C++** applications using **multithreading** and **multiprocessing**, analyzing throughput, latency, and scalability across compute-intensive workloads.
- Applied **TCP/IP** networking, synchronization primitives, and **OS fundamentals** to build reliable distributed workflows and benchmark performance under varying system loads.

PROJECTS

LearnedIndex - Benchmarking Learned Database Indexes (GitHub) — April 2026 – May 2026
- Built a **Java benchmark** comparing five index designs (sorted array, TreeMap, B+ tree, linear RMI, neural RMI) over **1M** sorted keys; the linear RMI achieved **6.3 ns/lookup – 14× faster** than binary search, **40× faster** than TreeMap.
- Showed model complexity does not equal performance: the neural RMI took **5–10× longer** to build with no lookup gain, while the linear RMI used **~625 KB vs. 12 MB** for the B+ tree and **86 MB** for TreeMap.

On-Device Industrial Incident Response Agent (GitHub) — February 2026
- Engineered an end-to-end AI agent combining a **LoRA fine-tuned Gemma model** with **RAG** to classify industrial incidents, extract structured operational data, and generate context-aware remediation guidance in real time.
- Built a production-oriented inference pipeline optimized for **edge deployment**, enabling **autonomous incident response** without reliance on cloud infrastructure.

PUBLICATIONS

IoT-Blockchain Sharing via Frequent Sender-Receiver Analysis, *IEEE I2CT* — 2024
Artificial Intelligence Application in Personalized Fintech, *Kepes Journal* — 2023